# exp262: does contacts-v1 need RoPE?

Issue [#262](https://github.com/Open-Athena/MarinFold/issues/262). contacts-v1 documents are self-describing and order-randomized — every statement carries its own `<pXXX>` coordinate and statement order is uniform noise — so the proposal is to delete RoPE and replace it with a width-3 causal *token smear*: mix the previous two tokens' embeddings into the current one, as in the nanogpt speedrun's smear module.

Phase 0 asks the trained default checkpoint whether there is anything to win, before any training is bought. Phase 1 is a 3-arm ablation at the exp232 1.5B architecture on a reduced token budget.

# Two is the tight window

Enumerating the contacts-v1 grammar: a token's (statement form, slot) role is NOT a function of the previous 1 token class — `(POS, POS)` is ambiguous between a contact's second argument and a residue statement's head — and IS a deterministic function of the previous 2. Lookback 3 adds nothing. Width-3 smear is the tight bound.

# What Phase 0 found

Three probes on `contacts-v1-exp199-cooldown-1.5B`, teacher-forced over ground-truth documents from the exp245 monomers, on a local A5000.

**The smear half is directly motivated.** Layer 1 holds two nearly pure previous-token heads (0.998 and 0.994 of their attention at offset 1) and a third splitting 0.68/0.19 across offsets 1 and 2 — a width-3 smear implemented in attention, at the cost of three heads.

**The NoPE half loses its mechanism but keeps its premise.** Co-referent retrieval is already distance-uniform out to 2048 tokens, so RoPE is not costing us long-range reach and the long-protein story is dead. But randomizing every exact cross-statement distance costs *less* than deterministically rescaling them at matched stretch, so the model genuinely is not reading those distances.

**What position is actually for: counting.** Keeping the position range fixed but letting two tokens share an id costs +1.23 nats, ten times any distance-randomizing intervention. The model uses position as an index, which is exactly what NoPE would take away — and exactly what the pre-registered stopping-behaviour guardrail was written to catch.

# The pilot: the two changes only work together

15M-parameter twins on 150M tokens of the real decontaminated corpus, three seeds per arm at each arm's own best learning rate: smear alone buys 0.034 nats, NoPE alone COSTS 0.085, and together they buy 0.157 — an interaction of 0.208. The gain is entirely in the structure section, the shuffled bag where the theory said it should land. Note this overturns the Phase 0 reading, which had the smear as the safe half and NoPE as the speculative one.

# The answer: no, and by half a model generation

Both full-budget 1.5B runs completed the full 145,200-step schedule. Final validation loss: control 2.9745, NoPE + width-3 smear 3.0021. The proposal is 0.0275 nats WORSE — about half the 0.053 nats the whole #75 to #117 generation was worth, in the wrong direction. Seventeen matched post-transition evals average +0.0166 and the final five agree to within 0.003, so this is not a marginal call.

The control finished 0.0173 better than exp232's run of the same recipe, which both confirms the newer marin pin is loss-neutral and bounds the run-to-run spread of the setup at about the size of the effect we were chasing.

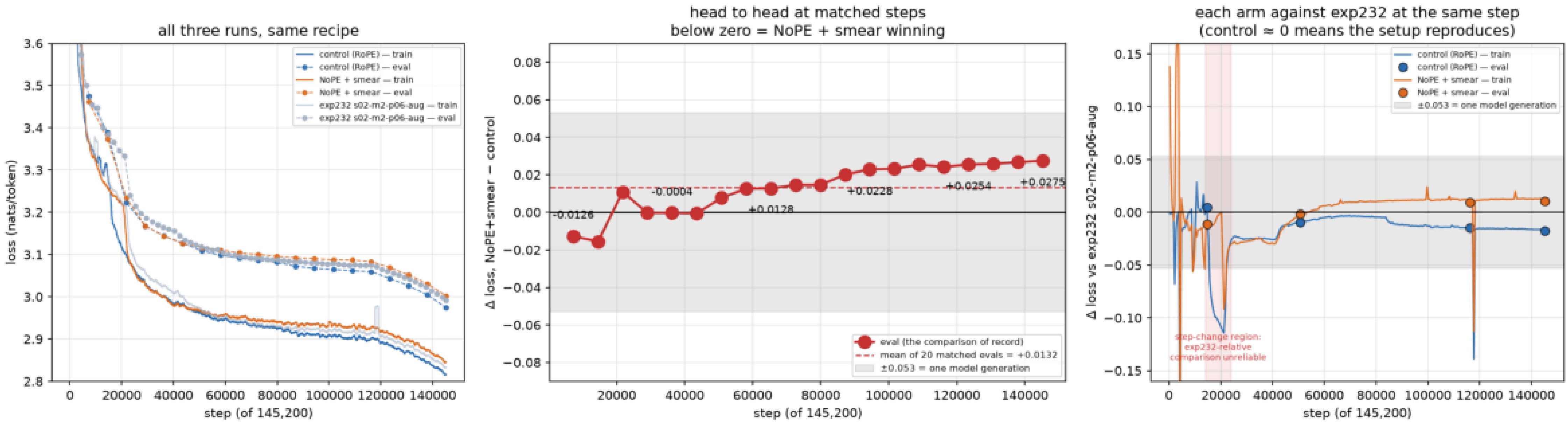
# Two things that outlast the negative result

The cheap proxies inverted the sign. A 15M pilot said -0.157 and a 10%-budget screen at the correct model size and data said -0.174; the answer was +0.0275. The screen is the worse offender: it ran the production model on production data and still got the sign wrong, because 14,520 steps ends before the transition below.

Every 1.5B run has one ~0.09-nat learning transition mid-training whose timing is NOT reproducible — the control's is near 15.5k and exp232's near 22k, on the same architecture, data and seed (review later found they also differed in embedding initialisation — a far smaller perturbation than architecture, but not nothing). That 6,500-step jitter is three times the effect under test. Any future mid-run architecture comparison at this scale needs seed replicates, and nobody had characterised this before.

# full_run_progress.png

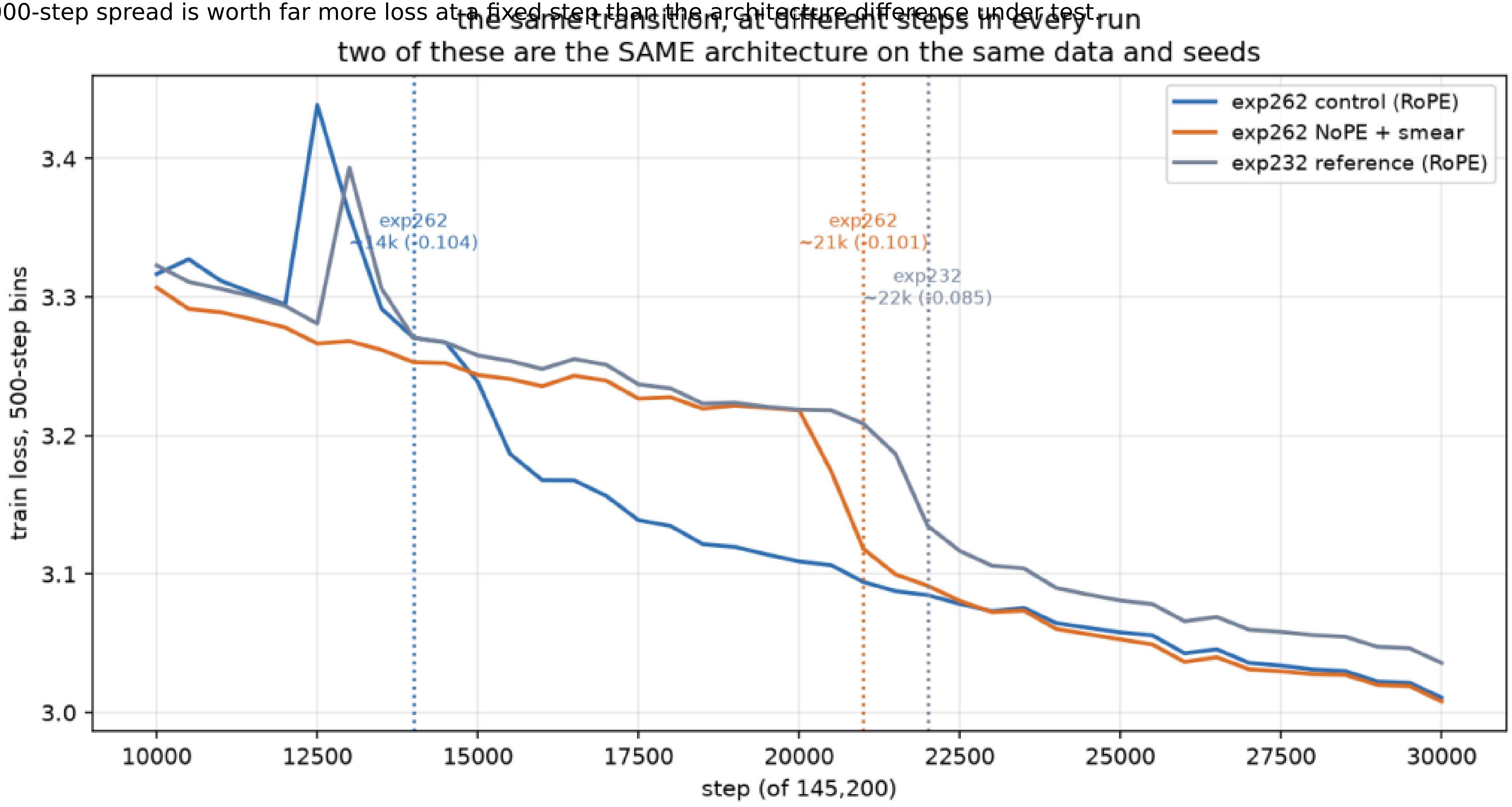
exp232's completed run is the common baseline. Middle panel: each arm minus exp232 at the same step — the control sits on zero, which is what shows the newer marin pin did not move the loss scale, and it is why the NoPE curve below zero can be read as architecture. Right: head to head where both have data; the control was preempted at step 21,535 and is catching up. Grey band is the 0.053 nats the #75 to #117 generation was worth. Middle is the comparison of record — both arms scored at identical steps on identical data; the small-scale pilot predicted about -0.16 and the mean of the matched evals is nowhere near it. Right: the red band is where all three runs have a step-change at different steps, so differencing across it measures timing rather than quality.




$ plot_full_runs.py

# loss_transition.png

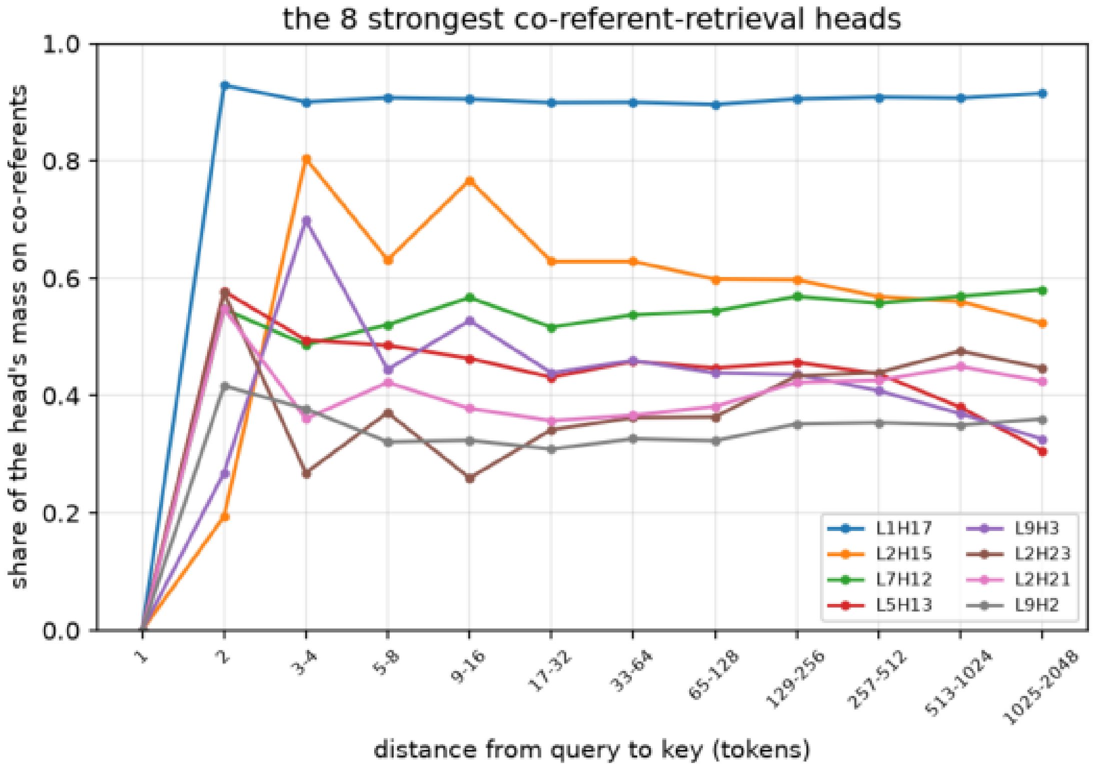
Each 1.5B run drops ~0.09 nats over ~1,500 steps somewhere mid-training. exp262's control transitions near 15.5k, its NoPE arm near 21k, exp232's reference near 21.5k — control and exp232 share architecture, data and seed. The timing is not reproducible, and its ~6,000-step spread is worth far more loss at a fixed step than the architecture difference under test.

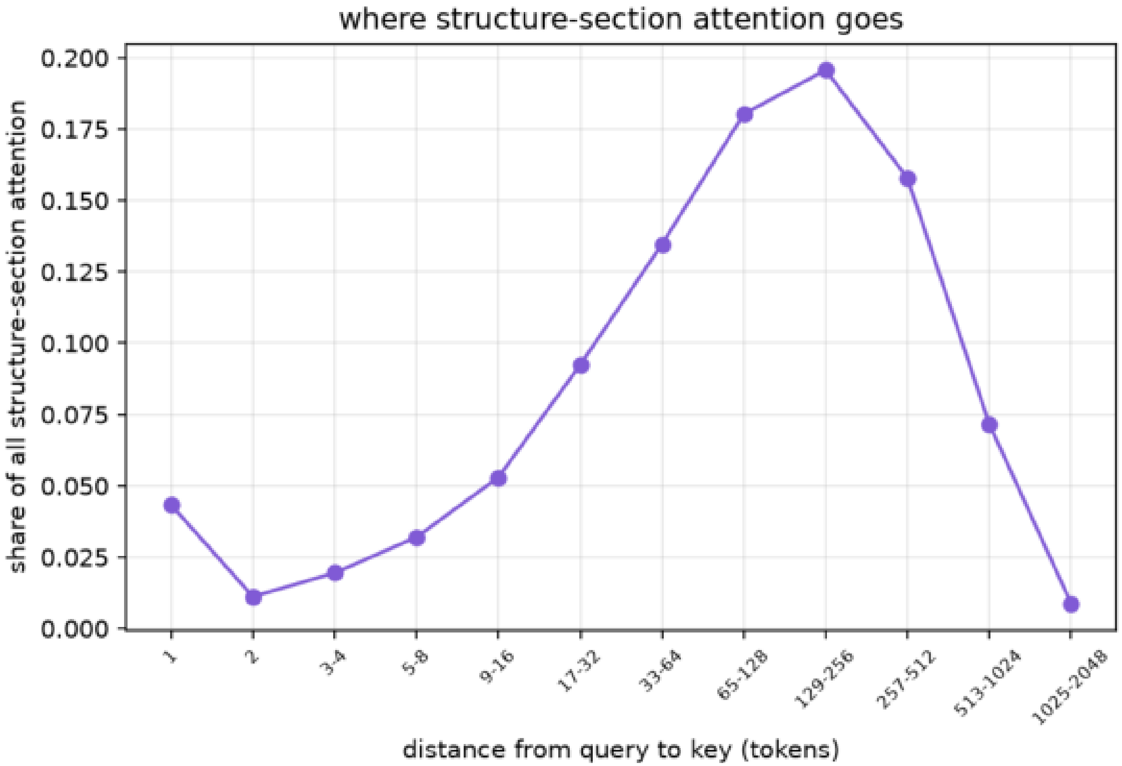



$ plot_transition.py

# phase0_coreferent_retrieval.png

Left: for heads that retrieve earlier mentions of the query's own residue index, the share of their attention on those co-referents, by distance. Flat — L1H17 holds 0.90-0.93 out to 2048 tokens. Retrieval is already distance-uniform, so RoPE costs us no reach and the mechanistic case for dropping it fails. Right: where structure-section attention goes.

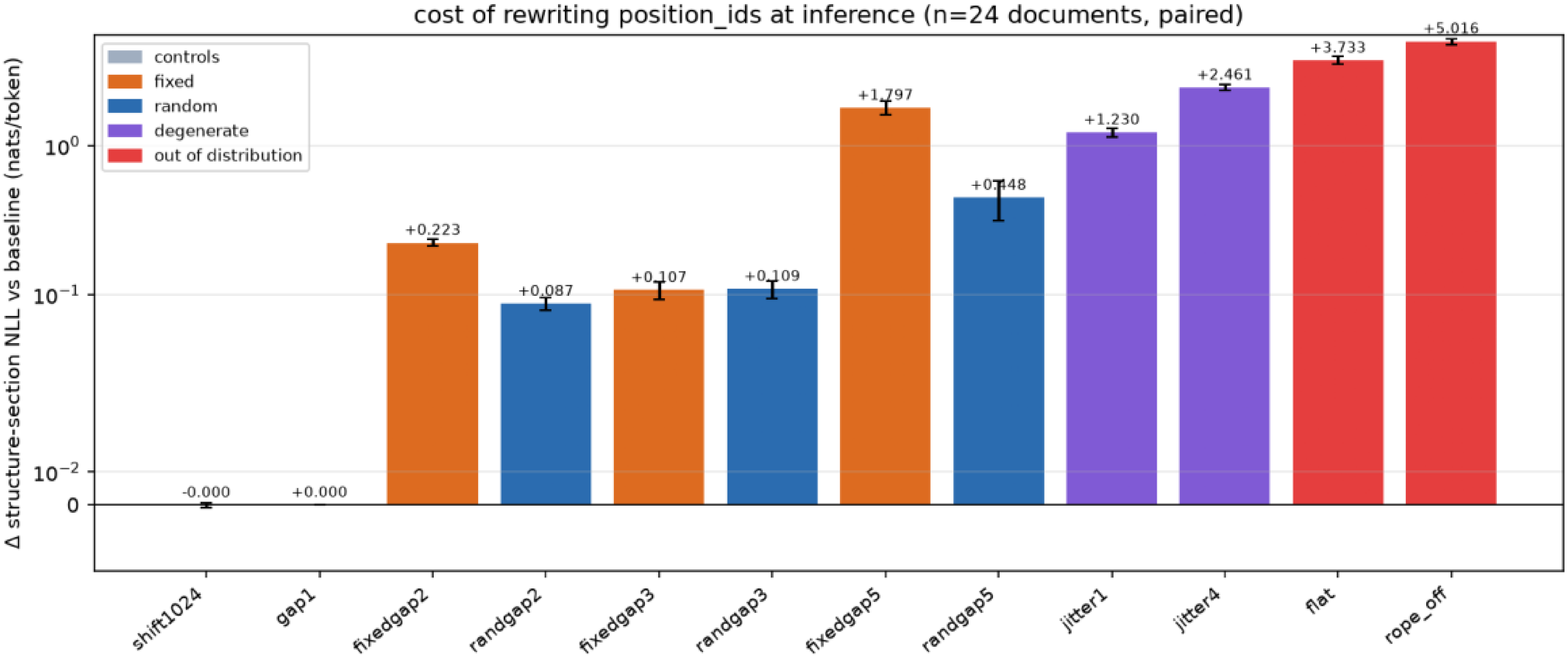



$ plot_phase0.py

# phase0_position_interventions.png

Paired change in structure-section NLL when position_ids are rewritten (n=24). fixedgapN and randgapN match on expected stretch and never repeat a position; only randgapN destroys exact cross-statement distances, and is never the worse of the pair — the model does not read them. jitterN instead repeats ids at fixed range and costs 10x more: position is an index.

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$ plot_phase0.py
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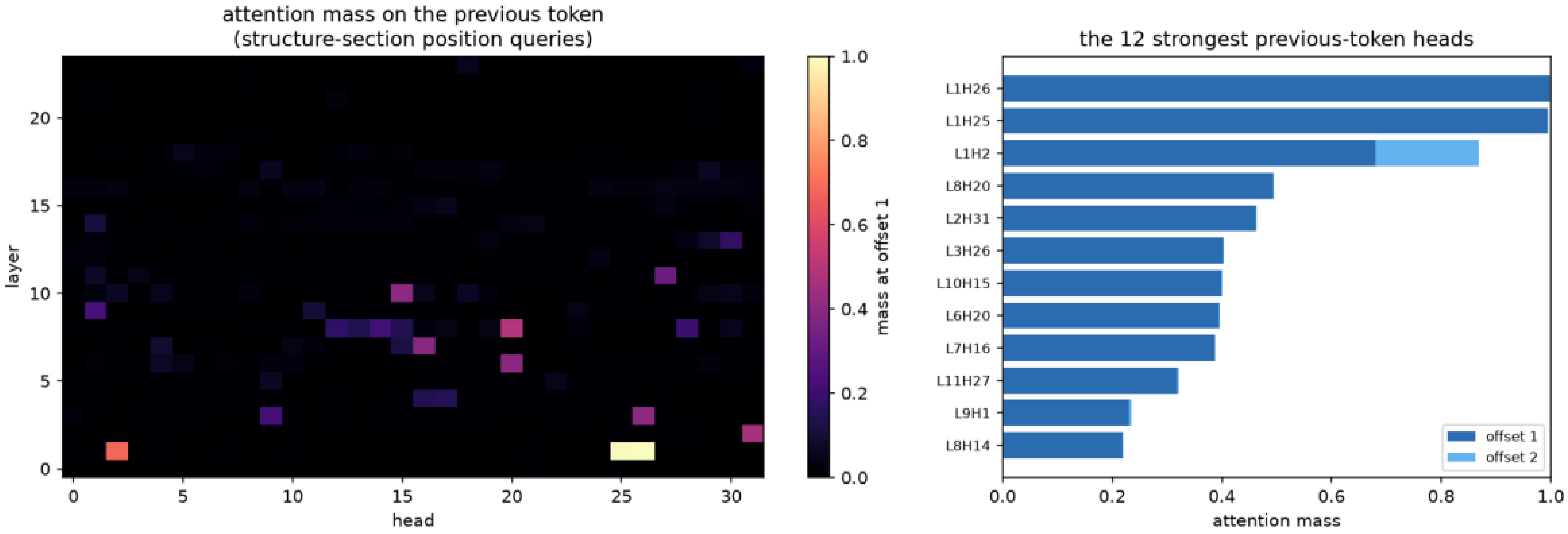
# phase0_previous_token_heads.png

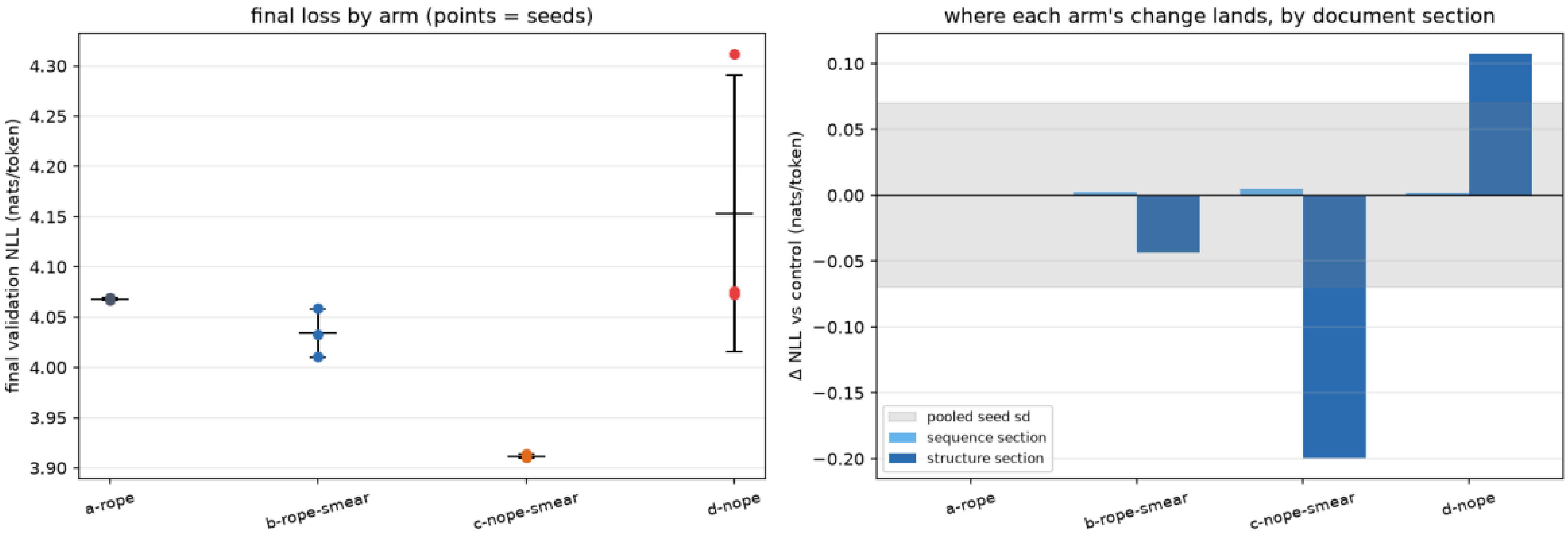
Attention mass at offset 1 per (layer, head), over structure-section position queries of 24 ground-truth documents. Layer 1 holds two near-pure previous-token heads (0.998, 0.994) and a third splitting 0.68/0.19 over offsets 1 and 2 — a width-3 smear built out of attention. Direct motivation for the smear half of #262.




$ plot_phase0.py

# pilot_arm_comparison.png

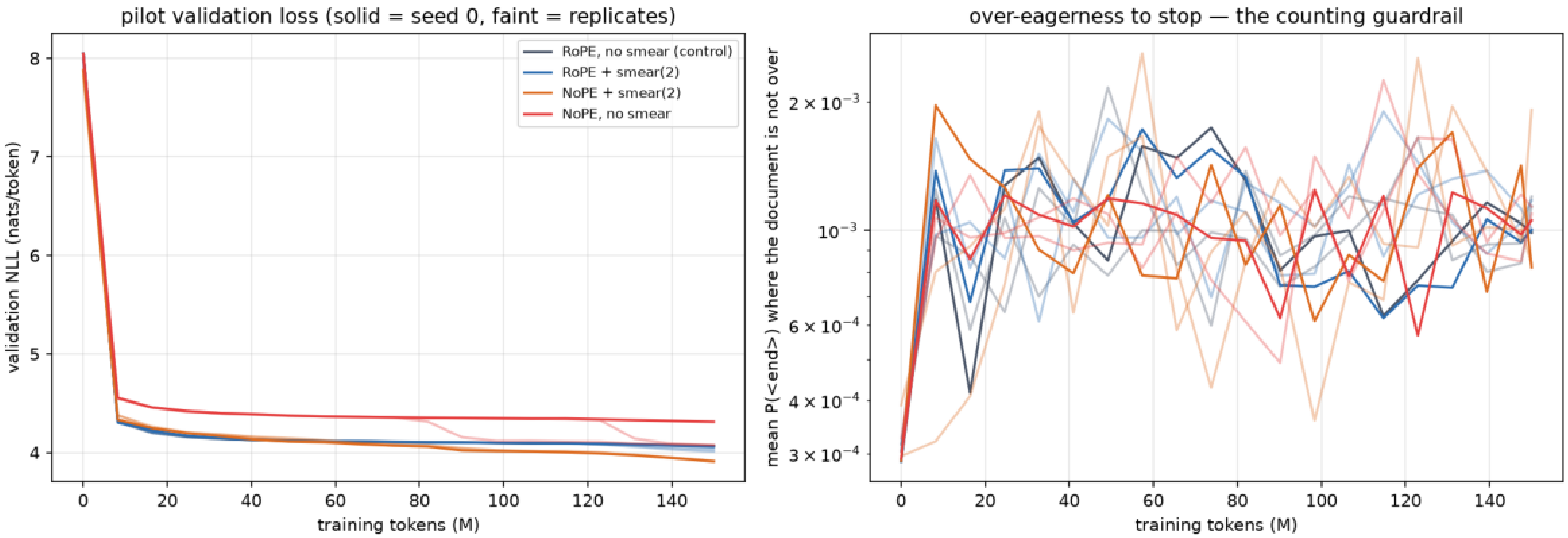
Left: final validation loss per arm, one point per seed, with the across-seed mean and spread. Right: each arm's gap to the control split by document section, against the pooled seed spread — the band is the pilot's resolution, and a bar inside it shows nothing.




$ analyze_pilot.py

# pilot_loss_curves.png

Local pilot: 15M-parameter twins of the exp232 Qwen3 on real decontaminated contacts-v1, one line per seed. Right panel is the guardrail Phase 0 asked for — the probability the model puts on <end> when the document is not finished, which is the counting job NoPE takes away.




`$ analyze_pilot.py`